

Guo, Siqu

+86-13021304295/ guosiqi2023@163.com/ Tianjin, China

<https://github.com/Guo-lab>

EDUCATION BACKGROUND

Tianjin University

09/2019-06/2023

- **Bachelor of Science** in Computer Science and Technology
- GPA: 3.84/4.0; Ranking: 5/139 (3%)

PUBLICATION

Ding, Y., & **Guo, S.** (2022, November). Conditional generative adversarial networks: introduction and application. In *2nd International Conference on Artificial Intelligence, Automation, and High-Performance Computing (AIAHPC 2022)* (Vol. 12348, pp. 258-266). SPIE.

Jin, D., Feng, B., **Guo, S.**, Wang, X., Wei, J., Wang, Z., Local-Global Defense Against Unsupervised Adversarial Attacks on Graphs. *Thirty-Seventh AAAI Conference on Artificial Intelligence*. In press.

ACADEMIC EXPERIENCE

Zero-shot Object Detection (Hitachi)

05/2022-09/2022

- Member, Advisor: Associate Professor Tang, Lingjia, University of Michigan
- Built a developed zero-shot object detection UI application from scratch using PyQt
- Managed various tasks prior to training the model, such as embedding the results of labels, modifying matrix dimensions using the CLIP model, and writing Shell scripts to automate processes
- Employed PDB to debug and reduced overheads in debugging time
- Analyzed the COCO dataset and KITTI dataset proficiently using pycoco for statistical analysis
- Utilized Netron and Tensorboard to visualize graphs and models in order to benchmark object detection models such as YOLO and FasterRCNN
- Implemented Tensorflow's Resnet50 object detection model as the baseline, and performed transfer learning on AlexNet, YOLOv5, and FasterRCNN

Theory and Applications of Graph Convolutional Network Adversarial Neural Networks

11/2021-09/2022

- Member, Advisor: Associate Professor Jin, Di, Tianjin University
- Graph Neural Network in Strategic Deployment
 - Uncovered the relationship between Graph Attention Transformer (GAT) and Graph Convolutional Network (GCN) applied in Deep Graph Infomax (DGI)
 - Learned about the theories proposed in *Unsupervised Adversarially Robust Representation Learning on Graphs*, combined DGI with GAT and introduced mutual information and unsupervised learning to DGI
- Robustness Analysis of Graph Neural Networks
 - Used Metattack and Nettack to manipulate data to obtain multiple datasets of malicious attack matrices
 - Reproduced and compared RoSA, Pro-GNN, PA-GNN, and other algorithms, and practically applied GraphCL and GraphSS to do defense analysis

Machine Learning and Data Science II - Development and Frameworks

07/2021-10/2021

- An online project-based course, Advisor: Professor Mark Vogelsberger, MIT
- Gained a comprehensive understanding of common development frameworks and how they can be used to implement data science and machine learning techniques
- Studied algorithms and architectures related to Machine Learning and Data Science, and completed a thesis on the Conditional Generative Adversarial Networks

Research on Campus Uncivilized Behavior Recognition Based on Deep Learning

06/2020-06/2021

- Member, Advisor: Associate Professor Yang, Liu, Tianjin University
- Explored the mechanics of human skeletal information, and learned about behavior recognition algorithms
- Attempted to improve the architecture of human skeleton recognition with federated learning
- Recorded and generated datasets from various locations on campus, performed supervised classification learning on a series of images and videos

CURRICULUM DESIGNS

Cache Simulator with L1 and L2 level	09/2022–11/2022
- Implemented a simple Cache Simulator with L1 level Cache and adjusted various cache eviction strategies such as LRU, LFU, WBWA, and WTNA in the report and visualize the result - Implemented advanced L2 Cache with an algorithm to effectively choose a victim cache entry - Analyzed the effects of various configurations, such as cache size, block size, and associability on the performance of the Cache Simulator	
Implementation of SQL Language Compiler (lexer and parser)	03/2022-04/2022
- Implemented the lexer and converted given SQL texts to the output of token sequences - Built a parser with a team that can establish FIRST set, FOLLOW set, and parsing tables for token sequences	
The Making of Racing Game and the Application Research of Light	10/2021-12/2021
Utilized Unity to create racing games and designed various in-game lighting rendering methods	
CPU Implementation and Genesys II FPGA On-board Testing Based on MIPS Architecture	09/2021-11/2022
- Completed the temu simulation, and utilized Qt to carry out the visualized UI - Implemented specific sections of the five-stage pipeline of CPU ISA based on Verilog - Designed some parts of the testing of MIPS instructions on the FPGA demo board - Uploaded and executed the scripts, completed the experiment report, and passed the oral defense	
Design of the Bilibili Database	09/2021-10/2021
Performed web crawling on the data of bilibili using BeautifulSoup, and built a web-based database using Flask	
Simulation of TCP Protocol in Computer Networks	08/2021-09/2021
- Simulated the entirety of the TCP Protocol with C++ - Decomposed and analyzed the three-way handshake protocol used by TCP, conducted program refactoring, and wrote the report to reflect on the findings - Verified the results of the TCP implementation by testing the packet loss rate, simulation transmission, etc.	

HONORS & AWARDS

Second Prize in APCMC (Top 15%), International Level	11/10/2021
Third Prize in MathorCup (Top 20%), National Level	04/20/2021
Bronze Award in 2020-2021 National College Students Algorithm Design and Programming Challenge Winter Competition, National Level	03/14/2021
Third Prize in National College Mathematics Competition	11/14/2020

ACTIVITIES

Member, Peiyang Chorus of Tianjin University	09/2020-09/2021
Member, Fresh Bamboo Painting and Calligraphy Association of Tianjin University	09/2019-09/2020
Debater, Debate Club, College of Intelligence and Computing of Tianjin University	09/2019-09/2020
Awarded Runner-up in the 2019 Debating College Freshman Competition	

COMPUTER SKILLS

Programming: C/C++ (Proficient), Matlab (Proficient), Python (Proficient), Markdown (Proficient), C# (Intermediate), CUDA (Intermediate), Shell (Intermediate), Golang (Intermediate), JAVA (Beginner), Rust (Beginner).
Operating Systems: macOS (Proficient), Windows (Proficient), Linux (Intermediate)
Web Servers and Databases: MySQL (Proficient), Pymysql (Intermediate)
Web Client: Flask (Proficient), JavaScript (Intermediate)
IDE: Anaconda (Proficient)
Applications: Microsoft Office (Proficient), Visual Studio (Proficient), ArcGIS (Proficient), Blender (Proficient), Notion (Intermediate), Qt Creator (Intermediate), Eclipse (Beginner), Xcode (Beginner), Arduino (Beginner)

TEST SCORES

TOEFL: 103 Reading 29, Listening 25, Speaking 23, Writing 26	01/2023
GRE: 329 verbal: 161/87%; quantitative 168/90%; AW 3.0/13%	01/2022